

Windows 10 VM and Sysmon Setup

Background

The first stage of the SOC automation lab was to prepare a Windows 10 endpoint that will later be monitored by Wazuh.

I created the Windows 10 virtual machine using Oracle VirtualBox. This machine will act as the endpoint in the SOC environment and will generate Windows security and system events.

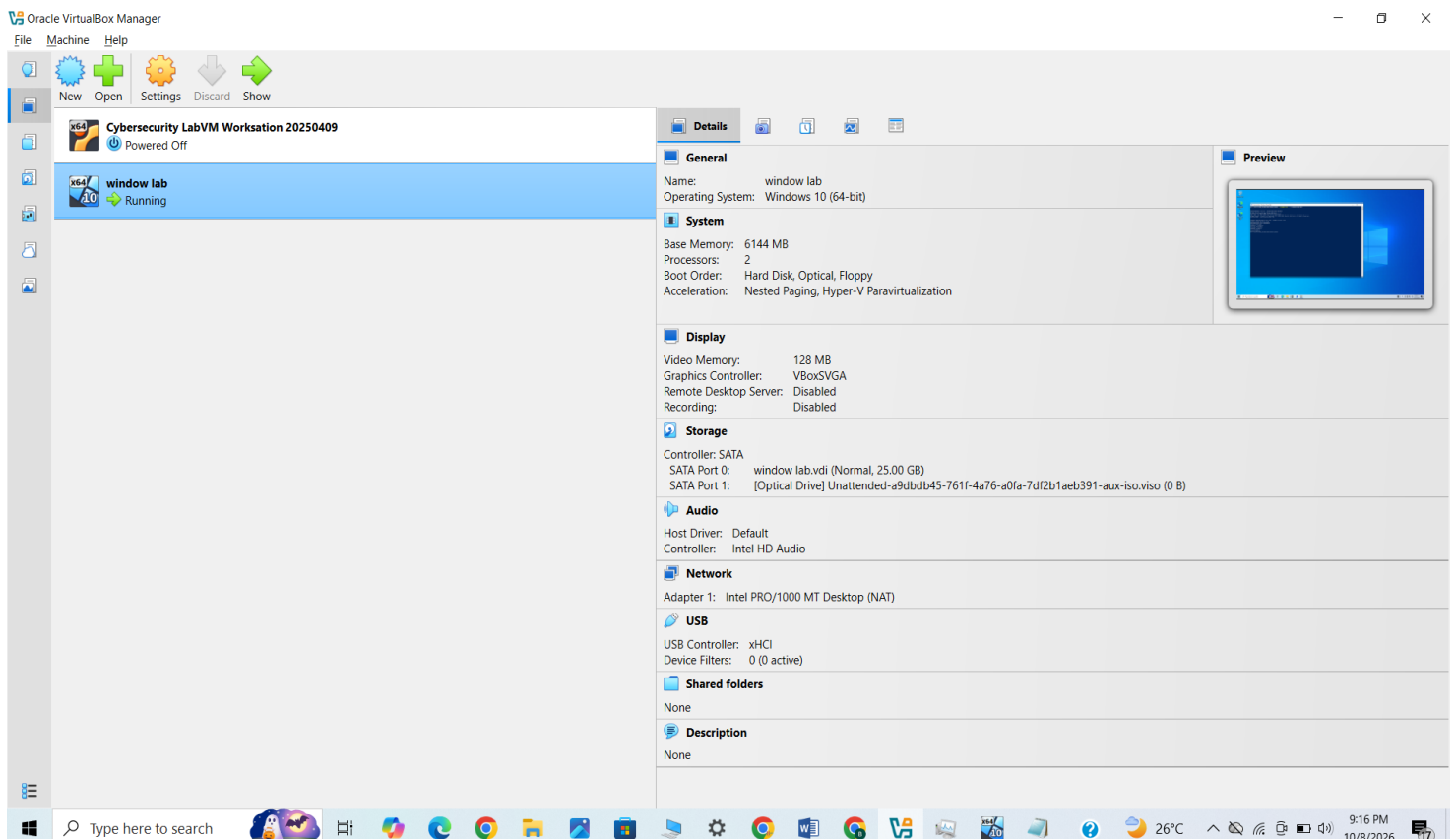
After setting up Windows 10, I installed **Sysmon (System Monitor)** to collect detailed system activity and generate additional telemetry that can be used for security monitoring and investigation.

Demonstration

The following screenshots show the Windows 10 virtual machine and the Sysmon installation and verification performed during the lab.

Screenshots

Windows 10 VM



Sysmon installation and configuration

```
Administrator: Windows PowerShell
PS C:\Windows\system32> cd .\downlevel\
PS C:\Windows\system32\downlevel> cd..
PS C:\Windows\system32> cd download
cd : Cannot find path 'C:\Windows\system32\download' because it does not exist.
At line:1 char:1
+ cd download
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Windows\system32\download:String) [Set-Location], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Windows\system32> cd C:\Users\shah_window\Downloads\Sysmon
PS C:\Users\shah_window\Downloads\Sysmon> dir

Directory: C:\Users\shah_window\Downloads\Sysmon

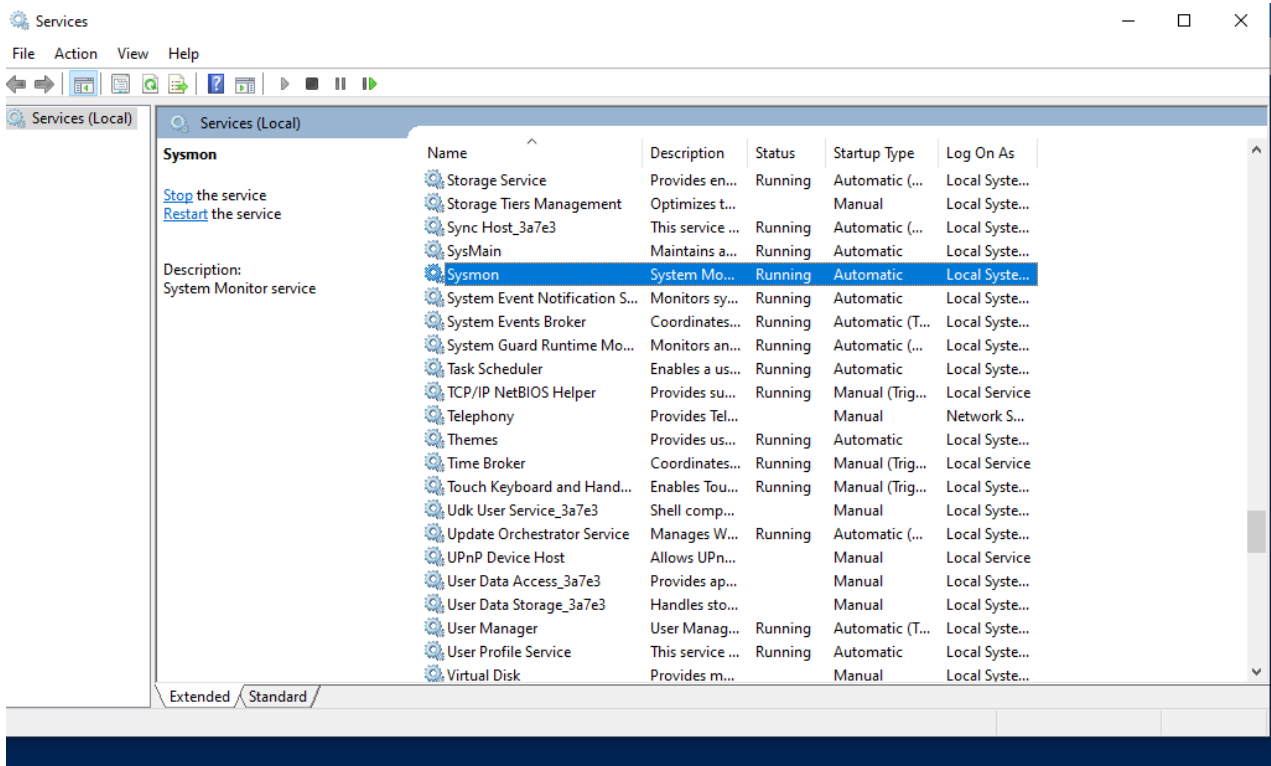
Mode                LastWriteTime         Length Name
----                -
-----
-----
9/10/2026   4:03 PM             7490 Eula.txt
-----
9/10/2026   4:06 PM          6162240 Sysmon.exe
-----
9/10/2026   4:06 PM          3203368 Sysmon64.exe
-----
9/10/2026   4:06 PM          3138824 Sysmon64a.exe
-a-----
10/8/2026   8:32 PM          123257 sysmonconfig.xml

PS C:\Users\shah_window\Downloads\Sysmon>
```

```
Administrator: Windows PowerShell
PS C:\Users\shah_window\Downloads\Sysmon> .\Sysmon.exe -i .\sysmonconfig.xml

System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon installed.
SysmonDrv installed.
Starting SysmonDrv.
```



Windows 10 Virtual Machine Setup

I created the Windows 10 virtual machine using **Oracle VirtualBox**.

VM Configuration

Configuration	Value
Operating System	Windows 10
Virtualization	Oracle VirtualBox
RAM	6 GB
CPU	2 cores
Virtual Disk	15 GB

The Windows 10 installation was performed manually using a Windows 10 ISO. During installation, I selected Windows 10 Pro and used the custom installation option.

This Windows 10 machine will be used as the endpoint for the SOC lab.

Sysmon

Sysmon (System Monitor) is a Windows system monitoring tool from Microsoft Sysinternals.

It runs as a Windows service and records detailed information about system activity, such as:

- Process creation
- Network connections
- File and system activity
- Process execution details
- Other security-relevant events

These events are stored in Windows Event Logs and can later be collected by security monitoring tools such as Wazuh.

For this SOC project, Sysmon is used to provide additional endpoint telemetry that can help with detecting and investigating suspicious activity.

Sysmon Installation

1. Download Sysmon

I downloaded Sysmon from the official Microsoft Sysinternals source.

After downloading the ZIP file, I extracted the Sysmon files on the Windows 10 machine.

2. Download the Sysmon Configuration

I also downloaded a Sysmon configuration file in XML format from a Git repository.

The configuration file defines which events Sysmon should monitor and log.

I placed the configuration file in the same directory as the extracted Sysmon files.

3. Open PowerShell as Administrator

I opened **PowerShell as Administrator** because installing Sysmon requires administrative privileges.

I then changed the directory to the location where the Sysmon files and configuration file were stored.

4. Check Sysmon Before Installation

Before installing Sysmon, I checked Windows Services and Windows Event Viewer.

There was no Sysmon service available, and the Sysmon event log was not present because Sysmon had not yet been installed.

5. Install Sysmon with Configuration

From the PowerShell window, I ran:

```
.\Sysmon64.exe -i SysmonConfig.xml
```

The `-i` option installs the Sysmon service and driver and applies the specified configuration file.

After running the command, Sysmon displayed its license agreement. I accepted the agreement and allowed the installation to continue.

6. Verify Sysmon Installation

After installation, I checked **Windows Services** again.

The Sysmon service was now available, confirming that the installation was successful.

I also opened **Event Viewer** and navigated to:

```
Applications and Services Logs  
→ Microsoft  
→ Windows  
→ Sysmon  
→ Operational
```

The Sysmon event log was now available and contained system telemetry generated by the Windows endpoint.

This confirmed that Sysmon was installed and collecting events successfully.

Result

The Windows 10 endpoint is now prepared with Sysmon for detailed system monitoring.

The Sysmon telemetry will later be used as part of the SOC workflow, where endpoint events can be collected and analyzed through Wazuh.